12-WEEK HYPERTROPHY PROGRAM

Second Gear

A four-day upper/lower program for the intermediate lifter. Every set, rep, and rest period explained - built from 40+ coaching and research sources, including where the experts disagree.

Intermediate · 4 days/week · 75-min sessions

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1. Who This Program Is For

This program is for you if:

- You have trained seriously for 1–3 years.
- You can commit to 4 gym days a week, about 75 minutes each.
- You want visible muscle growth, not just strength numbers.
- You have access to a full commercial gym — barbells, machines, cables.

This program is NOT for you if:

- You have trained less than a year — run a full-body 3-day program first; a little stimulus does a lot at that stage, and you will grow faster on less.
- You can only train 3 days a week — a different container fits that schedule better.
- You are returning from injury — clear it with a professional first.

> **What you will need:** a logbook (the included spreadsheet or printed sheets), a way to time 2-minute rests, and honesty about how close each set is to failure. That is the entire equipment list.

2. The Promise — and How This Program Is Different

Most programs hand you a table and say "trust me." This one does not.

Second Gear is built from 40+ sources — coaches and educators in the mold of Jeff Nippard, Andy Baker, J3 University, Westside Barbell, and John Meadows, plus the current hypertrophy research. Every choice in these pages has a reason attached, and where the experts genuinely disagree — how close to failure you should train, whether rear delts need direct work, whether calves respond to gym work at all — this book shows you **both positions** and tells you which one this program takes, and why.

You will also learn the skill that outlasts any program: reading your own logbook. Twelve weeks from now, the numbers — not soreness, not motivation — will tell you exactly what to do next. This program teaches you that loop.

> **What to expect:** 4 sessions a week — two upper, two lower. Every major muscle trained twice weekly with 10–13 hard sets per week, the growth zone for your training age. Progression runs in two blocks: Block 1 adds reps, Block 2 adds sets. Deloads at weeks 5 and 12. Simple levers, applied relentlessly.

3. How to Read This Program

RIR — Reps In Reserve

How many reps you could still do before failure. RIR 2 means you stopped 2 reps short. RIR 0 means you could not have completed another rep with good technique.

Hard (working) sets

The sets that count. Warm-up sets never count — do 1–2 light, easy ramp-up sets before the first exercise for each muscle, and only then do the numbered sets begin.

Rest

Standardized at 2 minutes everywhere. Not because 2:00 is magic — research supports 2–3 minutes for growth — but because you have 19–23 hard sets to fit into 75 minutes, and a rest you actually follow beats a perfect rest you do not.

Supersets

Marked pairs (A1/A2) are performed back-to-back with one shared 2-minute rest. The pairs are chosen so the two exercises do not compete — opposite muscles or different joints — so performance holds up on both.

Effort by exercise type — this matters

- **Heavy barbell compounds** (squat, bench, deadlift, RDL): stop at **RIR 1–3**, even on the last set. Grinding these to failure costs recovery you will pay for tomorrow, for little extra growth.
- **Machines and cables:** the **last set can go to failure**; earlier sets at RIR 1–2.
- **Isolations** (raises, curls, pushdowns): **failure is fine regularly** — the fatigue cost is local and small.

4. The Weekly Split

Day	Session
Mon	Upper A (chest-led)
Tue	Lower A (quad-led)
Wed	Rest
Thu	Upper B (back-led)
Fri	Lower B (hinge-led — deadlift day)
Sat / Sun	Rest

Why this layout:

- **Never more than 2 consecutive training days** — the mid-week rest keeps systemic fatigue from stacking up before each session.
- **Deadlifts on Friday**, in front of the weekend's two rest days — the most fatiguing lift gets the most recovery afterward.
- **Upper days alternate chest-led and back-led** so each pulling and pushing muscle gets a fresh start once a week.
- **Every muscle lands twice a week** — the consensus growth frequency among nearly all of our sources.

5. The Four Sessions

Upper A (Mon) — chest-led

#	Exercise	Sets × Reps	RIR	Rest
1	Barbell Bench Press	3 × 6–10	1–3	2 min
2	Chest-Supported Row	3 × 8–12	1–2	2 min
3	Incline DB Press	3 × 8–12	1–2	2 min
4	Lat Pulldown	3 × 8–12	1–2	2 min
5	DB Lateral Raise	3 × 12–15	0–1	2 min
6	Superset — A1 Rope Pushdown · A2 Hammer Curl	3 × 10–15 each	0–1	2 min shared

Coaching notes: Bench first while fresh — it is the session's one heavy free-weight compound and it stays 1–3 RIR. Incline press with a deep stretch at the bottom; the lengthened position is where growth is stimulated. Full stretch at the top of every pulldown. Side delts get direct work because pressing alone does not build them.

Substitutions: bench → machine chest press (joint-friendly) · chest-supported row → machine row or single-arm DB row · incline DB → incline Smith or machine press · pulldown → pull-up.

Lower A (Tue) — quad-led

#	Exercise	Sets × Reps	RIR	Rest
1	Back Squat	4 × 6–10	1–3	2 min
2	Romanian Deadlift	3 × 8–10	1–3	2 min
3	Seated Leg Curl	3 × 10–15	0–1	2 min
4	Standing Calf Raise	3 × 10–15	0–1	2 min
5	Cable Crunch	3 × 10–15	0–1	2 min

Coaching notes: Full-depth squats — depth is the stimulus, not a bonus. Seated leg curls over lying: the seated position trains the hamstring in its lengthened state. Pause in the stretched position at the bottom of every calf raise.

Substitutions: back squat → hack squat or Smith squat · RDL stays (it is the hinge slot) · cable crunch → roman-chair raises.

Upper B (Thu) — back-led

#	Exercise	Sets × Reps	RIR	Rest
1	Pull-Up (weighted when ready)	3 × 6–10	1–3	2 min
2	Machine Incline Press	3 × 8–12	1–2	2 min
3	Seated Cable Row	3 × 8–12	1–2	2 min
4	Pec Deck / Cable Crossover	2 × 10–15	0–1	2 min
5	Superset — A1 Cable Lateral Raise · A2 Rear-Delt Fly	3 × 12–15 each	0–1	2 min shared
6	Superset — A1 Overhead Triceps Extension · A2 Incline Curl	3 × 10–15 each	0–1	2 min shared

Coaching notes: Deep stretch at the bottom of every pull-up. The machine press slots in second because machines deliver tension with less fatigue once you are already worked. The overhead triceps extension is deliberate — the overhead angle is what loads the long head, the biggest part of the triceps.

Substitutions: pull-up → lat pulldown · pec deck → cable crossover · any curl/triceps pair keeps its overhead and stretch emphasis.

Lower B (Fri) — hinge-led

#	Exercise	Sets × Reps	RIR	Rest
1	Conventional Deadlift	3 × 6–8	1–3	2 min
2	Leg Press	3 × 10–12	1–2	2 min
3	Lying Leg Curl	3 × 10–15	0–1	2 min
4	Hip Thrust	3 × 8–12	1–2	2 min
5	Seated Calf Raise	3 × 10–15	0–1	2 min

Coaching notes: Deadlifts before the weekend. Feet low and narrow on the leg press — that is the quad stimulus position. Friday's calf raise is seated on purpose: a different angle from Tuesday's standing raises. Trap-bar deadlift is a fully equal substitution.

Substitutions: conventional deadlift → trap-bar deadlift.

6. Why This Volume — The Chapter Other Programs Don't Have

Every major muscle gets **10–13 hard sets per week**, spread across two sessions, with no more than about 7 sets of any muscle in one session. Here is where that comes from — and where our sources disagree.

The consensus

The growth zone for most muscles is 10–20 hard sets per week. Below 6–8 you grow slowly; above 20, returns shrink fast for most people. Within a single session, about 6 hard sets per muscle is where added sets stop adding much — which is why nothing here is smashed into one day.

The disagreement — and our position

One research-oriented camp argues that above 10–12 weekly sets the return is minuscule for most lifters; the coaching camp keeps clients growing at 15–20+. This program takes the **mid-band (10–13)** because you are training for growth with good recovery and a full four-day schedule — enough to grow briskly, not so much that your logbook drowns in fatigue.

Arms get "only" 6 direct sets each — on purpose. Direct biceps and triceps targets of 3–6 sets hold when arms are not the priority, because every one of your 12 back sets and 11 chest sets piles indirect arm work on top. Counting only the curls and pushdowns as "arm volume" misunderstands how volume adds up.

Where experts genuinely disagree — shown, not hidden

Rear delts. Camp 1: rear delts are overrated — proper back training (12 sets a week here) covers them. Camp 2: they are everyone's lagging head and need priority work. **Our position:** Camp 1, with 3 direct sets as insurance. If your rear delts visibly lag on this program, they are the first place to add sets in Block 2.

Calves. Camp 1: dedicated sets plus long stretch pauses work. Camp 2: calves respond to the load of daily life, and gym work mostly maintains them. **Our position:** a lean 6 sets with stretch pauses — if calves are your priority, that is the volume you would increase first, and you would be betting on Camp 1.

Failure training. Camp 1 (research-leaning): never grind compounds; isolations can fail. Camp 2 (one strong coaching voice): RIR targets are imprecise crutches; all-out sets build the most. **Our position:** Camp 1's split by exercise type (see chapter 3). If your logbook shows you grow on more all-out effort, that is a valid individual override — the numbers decide.

7. Progression — The Engine of the Program

Programs do not grow you; **progression** does. Second Gear runs two blocks, each with one lever. Changing two things at once means you cannot tell what worked.

Week 1 — calibration

Choose loads that land mid-rep-range at **RIR 3**. Deliberately conservative: you are buying room to climb. Log every set.

Block 1 (weeks 2–4) — add REPS

Same loads as week 1; add 1–2 reps per set each week. When all sets of an exercise hit the top of its rep range, add load next session (a small jump) and restart at the bottom of the range. This is **double progression** — the reps-first, load-second loop that runs the whole program.

> **Worked example:** Week 1: bench 185 lb × 8, 8, 8 @ RIR 3. Week 2: 185 × 9, 9, 9. Week 3: 185 × 10, 10, 9. Week 4: 185 × 10, 10, 10 → next session, 195 × 8, 8, 8.

Week 5 — deload

See chapter 8. It should feel easy — that means you timed it right.

Block 2 (weeks 6–11) — add SETS

Something must change after every block, so the lever switches from reps to sets: each week, add 1 set to one upper and one lower exercise — the ones still progressing. Cap 5 sets per exercise; at the cap, add to a different exercise instead. Reps continue double progression from week-5 levels.

> **Worked example:** Week 6: squat stays 4 × 6–10; leg press 3 × 10–12. Week 7: leg press becomes 4 × 10–12. Week 8 (after the deload): pulldown goes from 3 to 4 sets.

Week 12 — deload and re-test

After the final deload, repeat your week-6 loads on the four main lifts and compare logged reps. That comparison is the program's report card — and your evidence-based starting point for whatever comes next.

8. Deloads

Weeks 5 and 12: **halve every exercise's sets, keep the loads.** No new exercises, no intensity techniques. It should feel easy — that means you timed it right. If your light week still feels hard, that is confirmation you needed it.

Why halve sets and not load? Volume is this program's stressor; cutting the actual stressor recovers you, while dropping load just trains you lighter.

> **Symptom override — deloads do not wait for the calendar.** If you are beat up, soreness is lingering between sessions, warm-up weights feel heavy, or the pump has vanished, take the deload early regardless of what week it is.

9. Autoregulation — Reading the Logbook

Run these checks after every session:

- **Reps flat or down for 2 straight sessions on an exercise** → the prior volume was too high: drop 1 set from that exercise and rerun.
- **Reps moving** → change nothing. If you are progressing, it is the right amount of work. The most common intermediate mistake is changing a working program.
- **Stalled a full block** → use Block 2's set-addition lever on the stalled lift, or swap the exercise (any stall lasting multiple blocks, or reps falling below ~6 at your loads, is a swap signal).
- **Joint pain on any movement** → substitute the listed alternative that session. Pain is the one filter that outranks every rule in this book.

> *The principle underneath all four: the logbook is the judge. Not soreness, not the pump, not motivation.*

10. Missed Sessions

Life happens. Rules for keeping the week intact:

- **Missed one session:** do not double up — never stack two sessions in one day. Drop the missed session and resume the schedule; at 2×/week frequency every muscle still got one stimulus, and single-session frequency dips do not measurably stall growth.
- **Missed two:** convert to a full-body day — first exercise from each of the four sessions, 2 sets each, done in about 45 minutes. That is not "better than nothing" as a politeness; it is most of the stimulus.
- **A week off entirely** (travel, illness): resume where you left off, but repeat the previous week's loads at RIR 3 for the first sessions back. Do not punish yourself with a make-up week.

11. After Week 12

- **Rerun Block 2** with your re-test numbers if the program was still progressing — the simplest win.
- **Move to a new container** — a 3-day upper/lower/full program for busier seasons; the companion program, built on the same principles, ships soon.
- **Specialize** — if the re-test and the mirror agree one area lagged, that is when a specialization block earns its complexity. Not before.

Whatever you choose: carry the logbook habit with you. It outlasts every program, including this one.

Sources

Second Gear synthesizes roughly 40 coach-education and research-review sources, including Jeff Nippard-style science explainers (volume, frequency, rep ranges, rest intervals); Andy Baker coaching material (scheduling, volume cycling, deloads, heavy/light structure); J3 University split design; Westside/conjugate programming; John Meadows training philosophy; and current hypertrophy research on lengthened-position training, effort, and time-efficient sessions. A full source list with claim-level mapping is available at the product page.

This program is educational content, not medical advice. Consult a qualified professional before beginning any training program.